OTRS 11/#11/25 — Audrey & Isaiah

- We want to get a hilbert space $H \otimes K$ from two Hilbert spaces H & K.

Prop 3.2.1 $\otimes = \odot$

If H and K are Hilbert Spaces, then $H \otimes K$ is a pre-Hilbert space w.r.t. the inner product

$$(*) \quad \left\langle \sum_i h_i \otimes k_i , \sum_j h'_j \otimes k'_j \right\rangle = \sum_{i,j} \langle h_i, h'_j \rangle \langle k_i, k'_j \rangle.$$

Taking the Hilbert space closure (i.e. ~~adding in all the a~~ ensuring all Cauchy sequences converge w.r.t. this norm) we obtain an honest Hilbert space, still denoted $H \otimes K$
[book uses $H \odot K$ for the prehilbert sp.]

Proof WTS $\exists$ sesquilinear map

$$\langle \cdot, \cdot \rangle : H \otimes K \times H \otimes K \longrightarrow \mathbb{C}$$

satisfying above formula $(*)$. Let

$$(H \otimes K)^{*,c} = \{ A : H \otimes K \to \mathbb{C} \mid A \text{ conjugate linear (functionals)} \}$$

Now $\forall (h,k) \in H \times K$, $\exists$ functionals $g_h : H \to \mathbb{C}$, $v_i \mapsto \langle h, v_i \rangle$ and $g_k : K \to \mathbb{C}$, $w_i \mapsto \langle k, w_i \rangle$, $\therefore$ $\exists$ a tensor prod functional $f_{(h,k)} \in (H \otimes K)^{*,c}$

$$f_{(h,k)} : H \otimes K \longrightarrow \mathbb{C}$$
$$\sum_i v_i \otimes w_i \longmapsto \sum_i \langle h, v_i \rangle \langle k, w_i \rangle.$$

Now ~~the~~ we check that the assignment

$$\chi : H \times K \longrightarrow (H \otimes K)^{*,c}$$
$$(h,k) \longmapsto f_{(h,k)}$$

is bilinear:

(1) $\forall \alpha \in \mathbb{C}$, $\forall (h,k) \in H \times K$ we get

$$\alpha f_{(h,k)}\left(\sum_i v_i \otimes w_i\right) = \alpha \sum_i \langle h, v_i\rangle \langle k, w_i\rangle$$
$$= \sum_i \langle \alpha h, v_i\rangle \langle k, w_i\rangle = f_{(\alpha h,k)}(\Sigma \dots)$$
$$= \sum_i \langle h, v_i \rangle \langle \alpha k, w_i\rangle = f_{(h,\alpha k)}(\Sigma \dots)$$

$$\Rightarrow \alpha f_{(h,k)} = f_{(\alpha h,k)} = f_{(h,\alpha k)} \Rightarrow \alpha \Psi(h,k) = \Psi(\alpha h, k) = \Psi(h, \alpha k)$$

(2) ~~For~~ $(h,k)(h,k') \in H \times K$

$$f_{(h,k+k')}\left(\sum_i v_i \otimes w_i\right) = \sum_i \langle h, v_i\rangle \langle k+k', w_i\rangle$$
$$= \sum_i \langle h, v_i\rangle (\langle k, w_i\rangle + \langle k', w_i\rangle)$$
$$= \sum_i \langle h, v_i\rangle \langle k, w_i\rangle + \sum_i \langle h, v_i\rangle \langle k', w_i\rangle$$
$$= f_{(h,k)}\left(\sum_i v_i \otimes w_i\right) + f_{(h,k')}\left(\sum_i v_i \otimes w_i\right)$$

$$\Rightarrow \Psi(h, k+k') = \Psi(h,k) + \Psi(h,k')$$

(3) Analogously, $\Psi(h+h', k) = \Psi(h,k) + \Psi(h',k)$

$\therefore$ ~~By universal prop. of $\otimes$,~~ we furnish a linear map $M : H \otimes K \longrightarrow (H \otimes K)^{*,c}$. Then we define the desired sesquilinear form by

$$\langle \cdot, \cdot \rangle : (H \otimes K) \times (H \otimes K) \longrightarrow \mathbb{C}$$
$$\langle v, w \rangle = M(v) w \quad \forall v, w \in H \otimes K.$$

(2)

Our form is positive definite:

given an orthonormal ~~subset of vectors~~ vectors $\{e_i\} \subseteq H$, we calculate

$$\left\langle \sum_i e_i \otimes k_i, \sum_i e_i \otimes k_i \right\rangle = \sum_{i,j} \langle e_i, e_j \rangle \langle k_i, k_j \rangle$$

($\langle e_i, e_j \rangle = 0 \ \forall i \neq j$ b/c of orthogonality)

$$= \sum_i \|k_i\|^2 < \infty.$$

$0 <$

Inner prod = positive definite sesquilinear form.
So this is an inner prod. Now extend map to completion to get an inner prod. on $H \otimes K$. □

Now we want ONBs (you analysts are really asking for alot)

Prop 3.2.2 Let $\{v_i\}_{i \in I} \subseteq H$ and $\{w_j\}_{j \in J} \subseteq K$ are ONBs. Then $\{v_i \otimes w_j\}_{(i,j) \in I \times J} \subseteq H \otimes K$ is an ONB. Moreover, $\forall x \in H \otimes K$, $\exists!$ $\{k_i\} \subseteq K$ s.t. $x = \sum_i v_i \otimes k_i$, but this is no longer necessarily a finite sum, & convergence is in the norm topology.

forget to prove this, could talk through it

Proof. ~~A basis by a couple talks ago?~~ Check orthogonality/normality. Normality is easy:

$\langle v_i, v_i \rangle = 1 = \langle w_j, w_j \rangle \ \forall i,j$, so

$$\langle v_i \otimes w_j, v_i \otimes w_j \rangle = \langle v_i, v_i \rangle \langle w_j, w_j \rangle = 1$$

$$\Rightarrow \|v_i \otimes w_j\| = \sqrt{1} = 1.$$

Orthogonality

If $V_a \otimes W_b \neq V_c \otimes W_d$ then $a \neq c$ or $c \neq d$,

so $\langle V_a \otimes W_b, V_c \otimes W_d \rangle = \langle V_a, V_c \rangle \langle W_b, W_d \rangle$

$= \underbrace{0}_{} \text{ or } \underbrace{0}_{}$

0.

Thus, ONB.

(unless we also have to check lin indep. spanning set).

Spanning, easy to see. LI use

Prop 3.18. $\{x_1, \ldots, x_n\} \subseteq X$ l.i., $\{y_1, \ldots, y_n\} \in Y$ arbitrary, and $0 = \sum_{i=1}^{n} x_i \otimes y_i \in X \otimes Y$, then $y_1 = y_2 = \cdots = y_n = 0$

So if $0 = \sum_{i,j \in I \times J} v_i \otimes w_i$, since $\{v_i\}$ is l.i., then $w_j = w_{j'} \forall j, j' \in J$, so K is 1 dim'l, usually $\rightarrow\leftarrow$, otherwise, $H \otimes_{\mathbb{C}} \mathbb{C} \cong H$ and result is trivial (when k is 1 dim'l)

□

We saw tensor products of functionals earlier when furnishing a sesquilinear form on $H \otimes K$. How about tensor products of operators more generally?

If $S \in \mathcal{B}(H)$, $T \in \mathcal{B}(K)$, then there is an algebraic tensor map $S \odot T : H \odot K \to H \odot K$

Prop 3.2.3 If $S \in \mathcal{B}(H)$ and $T \in \mathcal{B}(K)$ are bdd lin ops on H and K, resp. then $\exists!$ bdd lin op $S \otimes T \in \mathcal{B}(H \otimes K)$ such that

$$S \otimes T (v \otimes w) = Sv \otimes Tw,$$

$\forall\, v \in H, w \in K$. Additionally, $\|S \otimes T\| = \|S\| \|T\|$.

Proof. We need:

Cor 3.1.10 $\{x_i\}_{i \in I} \subseteq X$ a basis, then $\forall v \in X \odot Y$, $\exists!\ \{y_i\}_{i \in I} \in Y$ s.t. $v = \sum_{i \in I} x_i \otimes y_i$.

Now consider $S = 1_H$. $\forall x \in H \odot K$ invoke the above and use Gram-Schmidt to obtain an orthonormal $\{e_i\} \subseteq H$ and unique $\{k_i\} \subseteq K$ s.t. $x = \sum_i e_i \otimes k_i$. Now we estimate

$$\begin{aligned} \| 1_H \odot T(x) \|^2 &= \| \sum_i e_i \otimes T(k_i) \|^2 \\ &= \sum_i \| T(k_i) \|^2 \\ &\leq \|T\|^2 \|x\|^2. \end{aligned}$$

Thus, $1_H \odot T$ has a unique extension to a bounded lin. op. on all of $H \otimes K$ (the completion of $H \odot K$), we denote it $1_H \otimes T$ and $\| 1_H \otimes T \| \leq \|T\|$, so $1_H \otimes T \in \mathcal{B}(H \otimes K)$. Analogously, one obtains $S \odot 1_K$ and extends to the completion $S \otimes 1_K \in \mathcal{B}(H \otimes K)$.

$\mathcal{B}(H \otimes K)$ is an algebra, (over $\mathbb{C}$) so we may define

$$S \otimes T := (S \otimes 1_K)(1_H \otimes T).$$

{ one may also check $S \otimes T]_{H \odot K} = S \odot T$

Evidently we obtain

$$\|S \otimes T\| \leq \|S\| \|T\| \quad \text{so} \quad S \otimes T \in \mathcal{B}(H \otimes K).$$

To show the reverse ineq., note that the norm is what we call a cross norm:

$\forall h \in H, \forall k \in K$, we have $\|h \otimes k\| = \|h\| \|k\|$.

Consequently, for unit vectors $h_n \in H$, $k_n \in K$ s.t. $\|S\| = \lim_{n \to \infty} \|S h_n\|$ and $\|T\| = \lim_{n \to \infty} \|T k_n\|$, then $h_n \otimes k_n \in H \odot K$ are still unit vectors and

$$\|(S \otimes T) h_n \otimes k_n\| = \|(S h_n) \otimes T(k_n)\|$$

$$= \|S h_n\| \|T k_n\| \xrightarrow{n \to \infty} \|S\| \|T\|.$$

$$\therefore \|S\| \|T\| = \|S \otimes T\|.$$

□

Now for some exercises.

Recall

$$\ell^2 = \left\{ (a_i)_{i \in \mathbb{N}} \;\middle|\; a_j \in \mathbb{C}, \sum_{i \in \mathbb{N}} |a_i|^2 < \infty \right\}$$

Ex(1). $\ell^2 \otimes \ell^2 \cong \ell^2$.

Pf. ℓ^2 is the unique (up to iso) seperable ∞ dim'l Hilbert space. So ℓ^2 has a countable schauder basis $\{e_i\}_{i\in\mathbb{N}}$ $e_i = (0,\dots,\underset{\uparrow i^{th}\text{ guy}}{1},0,\dots)$

Then $\{e_i \otimes e_j \mid i,j \in \mathbb{N}\}$ is a countable Schauder basis for $\ell^2 \otimes \ell^2$.

Thus, $\ell^2 \otimes \ell^2$ is seperable, and so $\ell^2 \otimes \ell^2 \cong \ell^2$.

□

Ex(2) $\ell^2 \otimes H \cong \ell^2(\mathbb{N}, H) = \left\{ (h_i)_{i\in\mathbb{N}} \mid h_i \in H, \sum_{i\in\mathbb{N}} \|a_i\|_H^2 < \infty \right\}$

Pf. It is not hard to chcek that the assignment

$$\varphi : \ell^2 \times H \longrightarrow \ell^2(\mathbb{N}, H)$$
$$((a_i), h) \longmapsto (a_i h)$$

is well-defined and bilinear, thus induces a $\mathbb{C}$-v.s. homomorphism

$$\varphi : \ell^2 \otimes H \longrightarrow \ell^2(\mathbb{N}, H)$$
$$(a_i) \otimes h \longmapsto (a_i h).$$

Bijectivity is unclear, ~~bdk.~~

According to Mikkel, ETS

① • orthonormal basis $\longrightarrow$ ONB $\Rightarrow$ preserves $\langle \cdot, \cdot \rangle$
$\Rightarrow$ isometry ✗

② • onto a dense subset:

②: $\{(h_i)_{i\in\mathbb{N}} \mid h_i = 0 \ \forall$ but finitely many $i\} = \mathcal{F}$
$\subseteq \ell^2(\mathbb{N}, \mathcal{H})$ is dense:

~~$\forall (h_i) \in \mathcal{F} \ \forall \varepsilon > 0 \ |B_\varepsilon(h_i) \cap \ell(\mathbb{N}, \mathcal{H})| \geq 2$~~

↳ easier $\mathcal{F} = \ell^2(\mathbb{N}, \mathcal{H})$ then

~~$\|(h_i) - (a_i)\| = \sqrt{\langle (h_i - a_i), (h_i - a_i) \rangle}$~~
~~$= \sum_i |h_i - a_i|^2 < \varepsilon$~~

now $\varphi(e_i, h_i)$ ~~$\equiv$~~ (h_i). So onto ctble dense subset.

① $(e_i) = (0, \dots, \underset{\text{ith guy}}{1}, 0, \dots)$ is an ONB of ℓ^2,
$\{f_i\}$ an ONB of $\mathcal{H}$, $\{e_i \otimes f_j \mid$ ~~ONB~~ $i, j\}$ ONB of $\ell^2 \otimes \mathcal{H}$.

$\varphi(e_i \otimes f_j) = (0, \dots, f_j, 0, \dots)$
ith guy

and if $\{f_j\}$ is an ONB of $\mathcal{H}$, then

$\delta_i : \mathbb{N} \longrightarrow \mathcal{H}$
$j \longmapsto \begin{cases} f_i & j = i \\ 0 & \text{else} \end{cases}$

$\{\delta_i\}$ is an ONB of $\ell^2(\mathcal{H})$.

$\therefore \ell^2 \otimes \mathcal{H} \cong \ell^2(\mathcal{H})$.

□